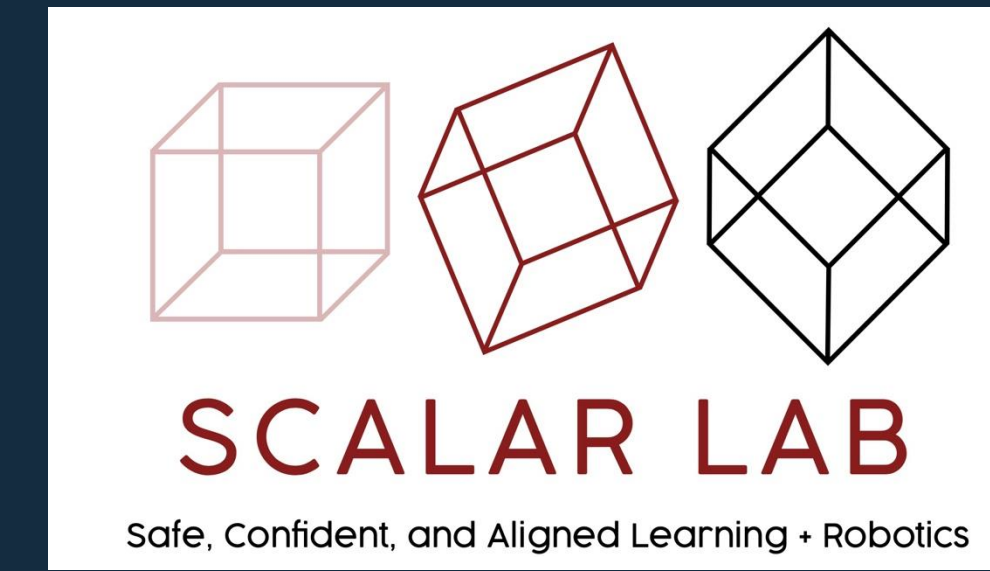


Safe Inference-Time Alignment via Lagrangian Reward Augmentation

Yaswanth Chittepu¹ · Ativ Joshi¹ · Sohini Chintala² · Scott Niekum¹

¹ University of Massachusetts Amherst ² Independent Researcher



Lagrangian Reward Augmentation (LARA) enables safety-aware inference-time alignment by turning an expected-cost constraint into a calibrated reward, without updating the language model.

The problem

- Changing safety requirements make repeated fine-tuning costly.
- Inference-time methods keep the model frozen, but steer it using a single score.
- That score has no direct connection to an explicit safety budget, so the reward–cost penalty is usually hand tuned.

Desired objective

$$\max_{\pi} \mathbb{E}_{\pi}[r] - \beta KL(\pi || \pi_o)$$

Subject to $\mathbb{E}_{\pi}[c] \leq \tau$

π_0 : reference model · r : reward / helpfulness · c : cost / harmfulness · τ : cost budget

Takeaways

- LARA turns an expected-cost budget into a calibrated reward for a frozen language model.
- The constrained problem reduces to one-dimensional convex calibration, with a finite-sample guarantee for estimating λ .
- With LARA, Best-of-N gives the strongest tradeoff among tested inference-time methods, approaching fine-tuned direct alignment baselines.

LARA: calibrate the safety penalty

Estimate the one-dimensional dual objective from calibration samples:

$$g(\lambda) = \beta \mathbb{E}_x \left[\log \mathbb{E}_{\pi_0} \left[\exp \left(\frac{r - \lambda c}{\beta} \right) \right] \right] + \lambda \tau$$

π_0 : reference model · β : KL weight · τ : cost budget

1. Sample responses and evaluate r and c
2. Minimize the estimated $g(\lambda)$ by binary search
3. Use $r - \hat{\lambda} c$ to score responses at inference time

Why binary search works

$$g'(\lambda) = \tau - \mathbb{E}_{\pi_{\lambda}^*}[c]$$

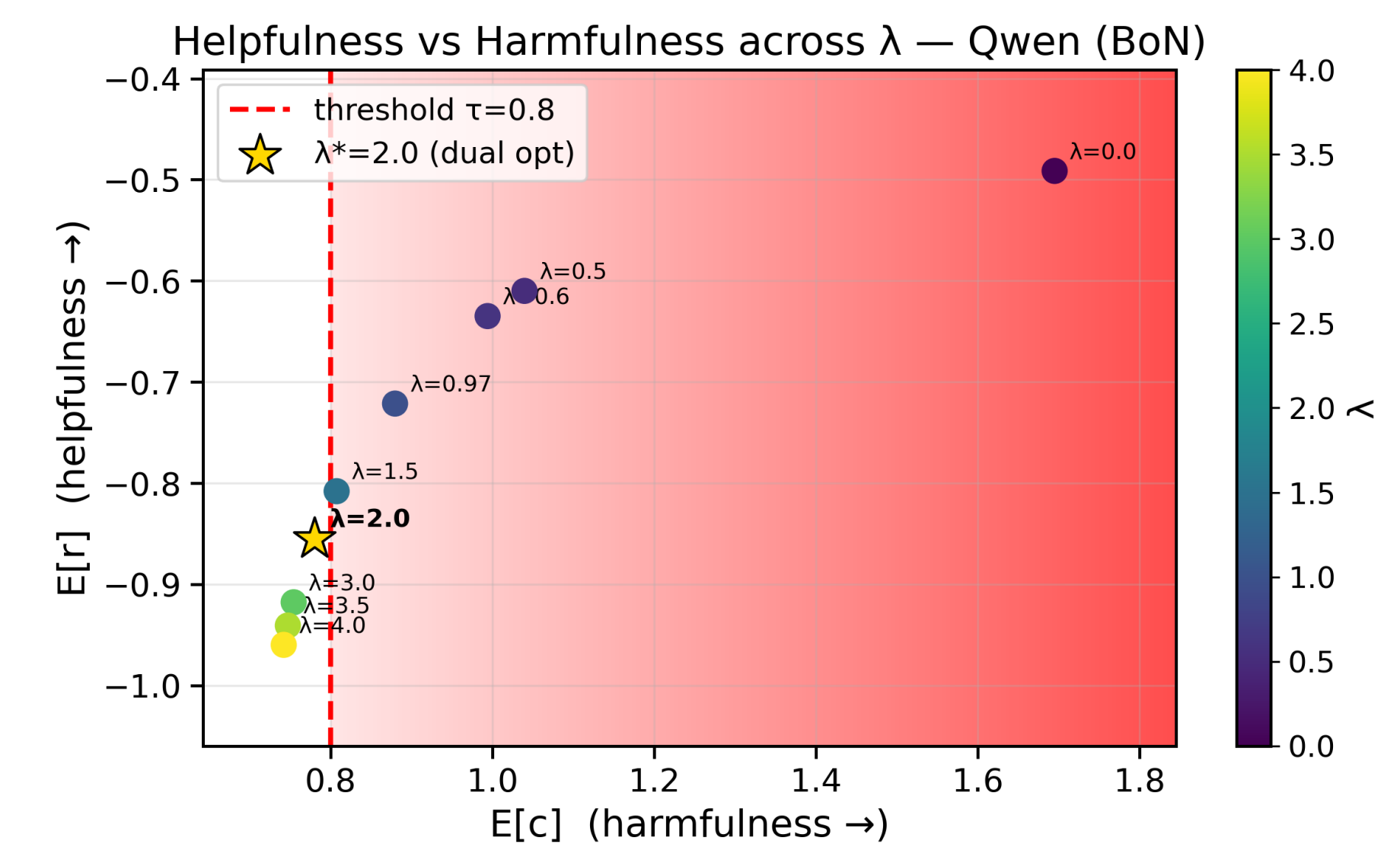
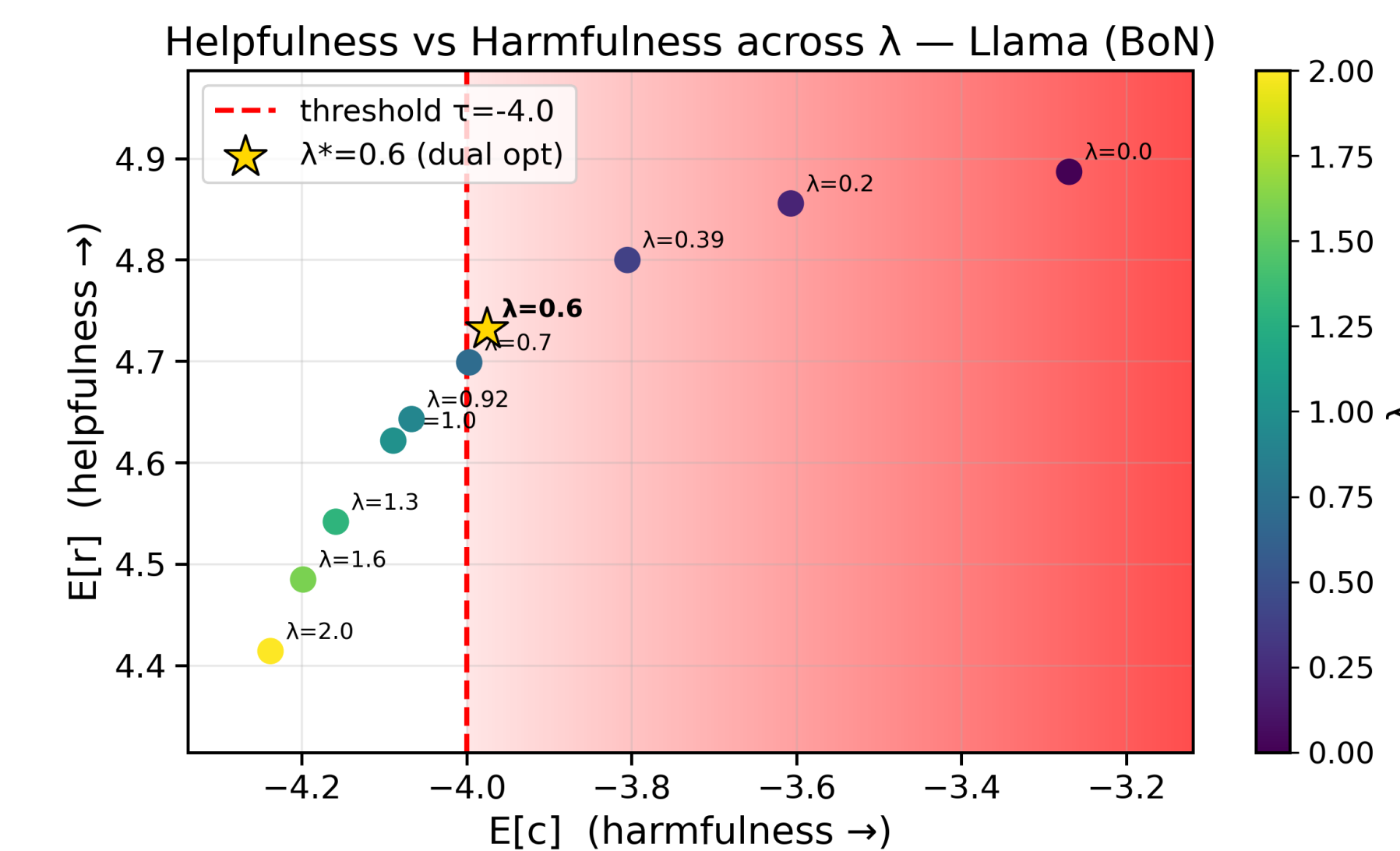
π_{λ}^* is the reference policy tilted by $r - \lambda c$: $\pi_{\lambda}^*(y|x) \propto \pi_0(y|x) \exp(\frac{r - \lambda c}{\beta})$

The expected cost decreases as λ increases. For an interior optimum, the expected cost reaches the budget τ .

This monotonicity of $g'(\lambda)$ enables efficient calibration of λ by binary search.

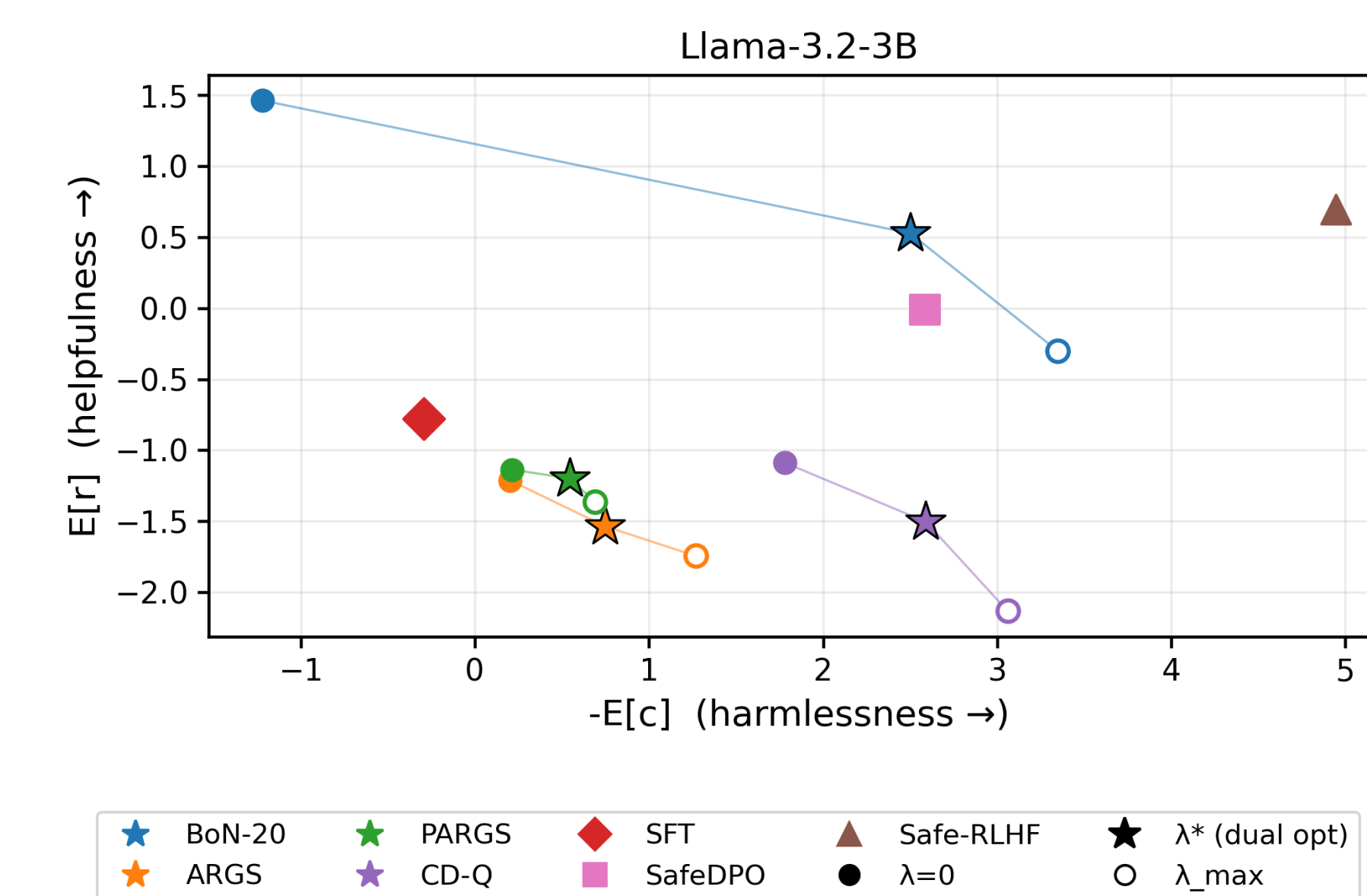
Scope: The ideal sequence-level policy has the constrained interpretation. Token-level decoders use the calibrated score as a heuristic.

Results

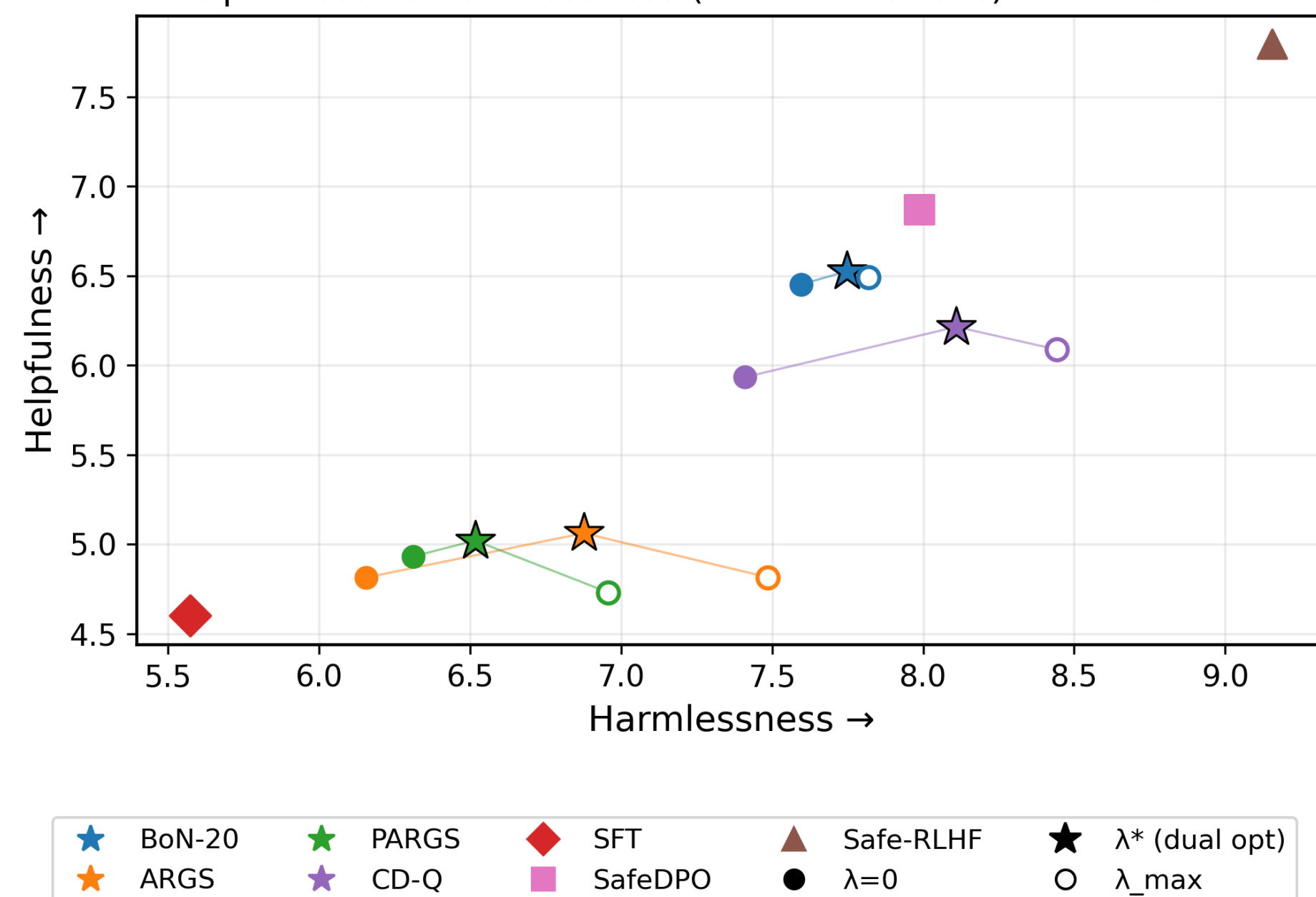


Calibrated λ reaches the cost threshold; larger penalties reduce helpfulness.

Helpfulness vs Harmlessness (Model Evaluations)



Helpfulness vs Harmlessness (GPT Evaluations) — Llama-3.2-3B



Best-of-N gives the strongest tradeoff among tested inference-time methods.